



Search for high-energy neutrino sources using IceCube data

Analysis of the 10-year point source sample

Master's Degree in Physics

Nicolò Favagrossa — Ali Arfat Haroon

July 4, 2026



UNIVERSITÀ
DEGLI STUDI
DI MILANO



Table of Contents

1 Introduction

► Introduction

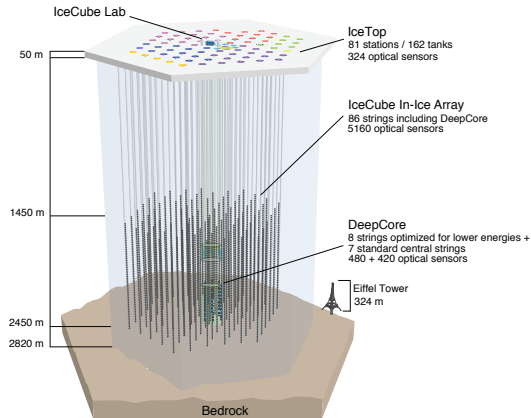


The IceCube Neutrino Observatory

1 Introduction

A Gigaton Cherenkov Detector

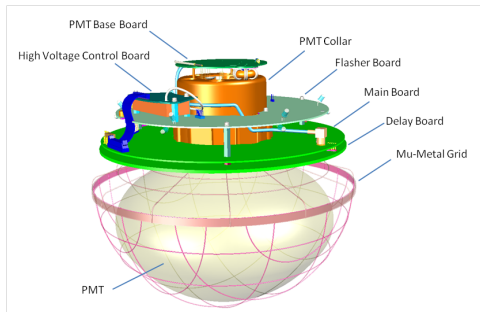
- **Location:** South Pole, Antarctica
- **Volume:** $\sim 1 \text{ km}^3$ of glacial ice
- **Depth:** 1450 m – 2450 m
- **Layout:** 86 strings, 5160 DOMs
- **Sub-detectors:**
 - *IceTop*: Surface array
 - *DeepCore*: Denser inner core





The Digital Optical Module (DOM)

1 Introduction



The Fundamental Detection Unit

- **Function:** Detects & digitizes Cherenkov light
- **Key Components:**
 - **PMT:** 10" photomultiplier
 - **Shield:** Mu-metal grid
 - **Mainboard:** Autonomous digitization & timestamping
 - **Vessel:** Pressure glass sphere
- **Data:** Sends digital signals to surface



Search for high-energy neutrino sources using IceCube data

Thank you for listening!
Any questions?